

Mordechai (Moti) Ben-Ari

Professor Emeritus
Department of Science Teaching
Weizmann Institute of Science

Education

- Ph.D. Tel Aviv University, Mathematics (Computer Science), 1981.
- M.Sc. Tel Aviv University, Mathematics (Computer Science), 1977.
- B.Sc., Mathematics, Massachusetts Institute of Technology, 1970.

Awards

- ACM Karl V. Karlstrom Outstanding Educator Award.
- ACM SIGAda Outstanding Ada Community Contribution Award, 2013.
- ACM Distinguished Educator, 2009.
- ACM SIGCSE Award for Outstanding Contributions to Computer Science Education, 2004.

Postdoctoral Fellows

1. Orni Meerbaum-Salant, 2010-12.
2. Michal Armoni, 2007–2009.

Ph.D. Students

1. Fatima Kaloti-Hallak. *The Effect of Robotics Activities on Students' Learning and Attitudes*, 2015. Joint supervisor: Michal Armoni.
2. Rivka Taub. *The Effect of Computer Science on Physics Learning in a Computational Environment*, 2014. Joint supervisor: Michal Armoni.
3. Jan Lönnberg. *Understanding and Debugging Concurrent Programs through Visualisation*, 2012.
4. Ronit Ben-Bassat Levy. *Teaching Computer Science with Animation: Attitudes and Ways of Experiencing*, 2009.
5. Niko Myller. *Collaborative Software Visualization for Learning: Theory and Applications*, 2009. Joint supervisor: Erkki Sutinen.
6. Noa Ragonis. *Teaching Object-Oriented Programming to Novices*, 2004.
7. Cecile Yehezkel. *A Visualization Environment for Computer Architecture*, 2004. Joint supervisor: Tommy Dreyfus.
8. Yifat Ben-David Kolikant. *Understanding Concurrency: The Process and the Product*, 2003.

M.Sc. Students

1. Mor Friebroon Yesharim. *What do elementary-school students learn in computer science classes with robotics?*, 2018.
2. Fatima Kaloti-Hallak. *Learning Programming Concepts Using Scratch at the Middle-School Level*, 2010. Joint supervisor: Michal Armoni.
3. Rivka Taub. *CS Unplugged and Middle-School Students' Views, Attitudes, and Intentions Regarding CS*, 2010. Joint supervisor: Michal Armoni.
4. Gil Ebel. *The Effect of Program Visualization on the Attention-Directing Characteristics of the Learner*, 2006.

5. Shmuel Schwartz. *Using State Diagrams for Understanding Correctness in Concurrent Programming*, 2006.
6. Andrés Moreno García. *The Design and Implementation of Intermediate Codes for Software Visualization*, 2004. Joint supervisor: Erkki Sutinen.
7. Yevgeniya Kulikova. *Roles of Variables in Teaching Functional Programming*, 2004. Joint supervisor: Jorma Sajaniemi.
8. Maxim Mozgovoy. *Concurrent Program Verifier: A Tool for Teaching Concurrent Programming*, 2004. Joint supervisor: Erkki Sutinen.
9. Niko Myller. *The Fundamental Design Issues of Jeliot 3*. Department of Computer Science, 2004, Joint supervisor: Erkki Sutinen.
10. Tzipora Yeshno. *Teaching an Explicit Conceptual Model As a Means to Improve the Work with Computer Applications*, 2003.
11. Ronit Ben-Bassat Levy. *The Use of Animation as an Educational Tool*, 2002.
12. Yekaterina Sedletzky. *Formal Verification of Distributed Algorithms*, 2000. Joint supervisor: Amir Pnueli.
13. Yakov Persky. *Re-engineering a Concurrency Simulator*, 1999.
14. Yoav Tsruya. *A Distributed Programming Environment in Ada95/Java*, 1998.
15. Noa Ragonis. *Introduction to Expert Systems: Development and Evaluation of a Computer Science Curriculum*, 1997. Joint supervisors: Ehud Shapiro, Zehava Scherz.
16. Ophira Statman. *A Prolog Proof Checker for Temporal Logic*, 1986.

List of Publications

Textbooks

1. M. Ben-Ari. *Principles of Concurrent Programming*. Prentice-Hall International, 1982.
2. M. Ben-Ari. *Principles of Concurrent and Distributed Programming*. Prentice-Hall International, 1990. Second Edition, Addison-Wesley, 2006.
3. M. Ben-Ari. *Mathematical Logic for Computer Science*. Prentice-Hall International, 1992. Second Edition, Springer, 2001. Third Edition, Springer, 2012.
4. M. Ben-Ari. *Understanding Programming Languages*. John Wiley, 1995.
5. M. Ben-Ari. *Ada for Software Engineers*. John Wiley & Sons, 1998. Second Edition with Ada 2005, Springer, 2009.
6. M. Ben-Ari. *Just a Theory: Exploring the Nature of Science*. Prometheus, 2005.
7. M. Ben-Ari. *Principles of the Spin Model Checker*, Springer, 2008.
8. M. Armoni, M. Ben-Ari. *Computer Science Concepts in Scratch*, Weizmann Institute of Science, 2010.
9. M. Ben-Ari, F. Mondada. *Elements of Robotics* (open access), Springer, 2018.
<https://link.springer.com/book/10.1007/978-3-319-62533-1>.
10. M. Ben-Ari. *Mathematical Surprises* (open access), Springer, 2022.
<https://link.springer.com/book/10.1007/978-3-031-13566-8>.

11. M. Ben-Ari. *The Geometry of Ellipses and Planetary Orbits* (open access), Springer, 2022.
<https://link.springer.com/book/10.1007/978-3-032-26272-1>.

Refereed Journals

12. M. Ben-Ari. Ianov pushdown schemes are contained in boolean recursive schemes. *Acta Informatica* 10(1977), 117–125.
13. M. Ben-Ari. On transposing large $2^n \times 2^n$ matrices. *IEEE Transactions on Computers* C28 (1979), 72–75.
14. M. Ben-Ari. Why you should not time-share. *Software—Practice and Experience* 9(1979), 339–340.
15. M. Ben-Ari. A simplified proof that regular resolution is exponential. *Information Processing Letters* 10(1980), 96–98.
16. M. Ben-Ari. Comments on “Tautology testing with a generalized matrix reduction method”. *Theoretical Computer Science* 11(1980), 341.
17. M. Ben-Ari. Cheap concurrent programming. *Software—Practice and Experience* 11(1981), 1261–1264.
18. M. Ben-Ari, J.Y. Halpern, A. Pnueli. Deterministic propositional dynamic logic: Finite models, complexity, and completeness. *Journal of Computer and System Sciences* 25(2), 1982, 402–417.
19. M. Ben-Ari, Z. Manna, A. Pnueli. The temporal logic of branching time. *Acta Informatica* 20, 1983, 207–226.
20. M. Ben-Ari. Algorithms for on-the-fly garbage collection. *ACM Transactions on Programming Languages and Systems* 6(3), 1984, 333–344.
21. M. Ben-Ari. How to solve the Santa Claus problem. *Concurrency: Practice and Experience* 10(6), 1998, 485–496.
22. M. Ben-Ari. Constructivism in computer science education. *Journal of Computers in Mathematics and Science Teaching* 20(1), 2001, 45–73.
23. M. Ben-Ari. Interactive execution of distributed algorithms. *ACM Journal on Educational Resources in Computing*, 1(2), 2001.
24. M. Ben-Ari. Theory-guided technology in computer science. *Science & Education* 10(5), 2001, 477–484.
25. R. Ben-Bassat Levy, M. Ben-Ari, P.A. Uronen. The Jeliot 2000 program animation system. *Computers & Education* 40(1), 2003, 1–15.
26. M. Ben-Ari. The NOMA of Yishayahu Leibowitz. *Science & Education* 12(7), 2003, 719–723.
27. M. Ben-Ari. On random numbers and design. *Science & Education* 13(3), 2004, 235–241.
28. M. Ben-Ari. Situated learning in computer science education. *Computer Science Education* 14(2), 2004, 85–100.
29. M. Ben-Ari. Situated learning in this high-technology world. *Science & Education* 14(3-5), 2005, 367–376.
30. N. Ragonis, M. Ben-Ari. A long-term investigation of the comprehension of OOP concepts by novices. *Computer Science Education* 15(3), 2005, 203–221.

31. M. Ben-Ari, T. Yeshno. Conceptual models of software artifacts. *Interacting with Computers* 18 (6), 2006, 1336–1350.
32. J. Sajaniemi, M. Ben-Ari, P. Byckling, P. Gerdt, Y. Kulikova. Roles of variables in three programming paradigms. *Computer Science Education* 16(4), 2006, 261–279.
33. C. Yehezkel, M. Ben-Ari, T. Dreyfus. The contribution of visualization to learning computer architecture. *Computer Science Education* 17(2), 2007, 117–127.
34. Y. Ben-David Kolikant, M. Ben-Ari. Fertile zones of cultural encounter. *Journal of the Learning Sciences* 17(1), 2008, 1–32.
35. R. Ben-Bassat Levy, M. Ben-Ari. Adapting and merging methodologies in doctoral research. *Computer Science Education* 19(2), 2009, 51–67.
36. M. Armoni, M. Ben-Ari. The concept of nondeterminism: Its development and implications for education. *Science & Education* 18(8), 2009, 1005–1030. Reprinted in: *inroads: SIGCSE Bulletin* 41(2), 2009, 141–160.
37. N. Myller, R. Bednarik, M. Ben-Ari, E. Sutinen. Extending the engagement taxonomy: Software visualization and collaborative learning. *ACM Transactions on Computing Education*, 2009, 7:1–7:27.
38. M. Ben-Ari, R. Bednarik, R. Ben-Bassat Levy, G. Ebel, A. Moreno, N. Myller, E. Sutinen. A decade of research and development on program animation: The Jeliot experience. *Journal of Visual Languages and Computing*, 22(5), 375–384, 2011.
39. R. Taub, M. Ben-Ari, M. Armoni. CS Unplugged and middle-school students' views, attitudes, and intentions regarding CS. *ACM Transactions on Computing Education*, 12(2), 8:1–8:29, 2012.
40. O. Meerbaum-Salant, M. Armoni, M. Ben-Ari. Learning computer science concepts with Scratch. *Computer Science Education*, 23(3), 2013, 239–264.
41. M. Armoni, O. Meerbaum-Salant, M. Ben-Ari. From Scratch to “Real” Programming. *ACM Transactions on Computing Education*, 14(4), article 25, 2015.
42. R. Taub, M. Ben-Ari, M. Armoni. The effect of computer science on physics learning in a computational science environment. *Computers & Education*, 87, 10–23, 2015.
43. F. Kaloti-Hallak, M. Armoni, M. Ben-Ari. The effectiveness of robotics competitions on students' learning of computer science, *Olympiads in Informatics* 9, 89–112, 2015.
44. M. Ben-Ari. LearnSAT: A SAT solver for education. *Journal of Open Source Software*, 3(24), 639, 2018. <https://doi.org/10.21105/joss.00639>.
45. M. Friebroon-Yesharim, M. Ben-Ari. Teaching computer science concepts through robotics to elementary school children. *International Journal of Computer Science Education in Schools* 2(3), 2018.
46. R. Taub, M. Ben-Ari, M. Armoni. Physics conceptual understanding in a computational science course. *Journal of Computational Science Education* 9(2), 2018.
47. F. Kaloti-Hallak, M. Armoni, M. Ben-Ari. The effect of robotics activities on learning the engineering design process. *Informatics in Education* 18(1), 105–129, 2019.

Conference Proceedings

48. M. Ben-Ari, Z. Manna, A. Pnueli. The temporal logic of branching time. *Eighth ACM Symposium on Principles of Programming Languages*. Williamsburg, VA, 1981, 164–176.

49. M. Ben-Ari, J.Y. Halpern, A. Pnueli. Finite models for deterministic propositional dynamic logic. *Eighth International Colloquium on Automata, Languages and Programming*. Haifa, Israel, 1981, 249–263.
50. M. Ben-Ari. On-the-fly garbage collection: New algorithms inspired by program proofs. *Ninth International Colloquium on Automata, Languages, and Programming*. Århus, Denmark, 1982, 14–22.
51. M. Ben-Ari. Reverse engineering into Ada. *Workshop on Software Methodologies in Ada*, Tel-Aviv, Israel, 1983.
52. H. Yashinsky, M. Ben-Ari. Ada for military microcomputers. *Second Israel Conference on Computer Systems Engineering and Software Engineering*. Tel-Aviv, Israel, 1987 (in Hebrew).
53. M. Ben-Ari. Ada requirements for small real-time systems. *Third International Workshop on Real-Time Ada Issues*. Nemaquin Woodlands, PA, 1989, 159–165.
54. M. Ben-Ari. Experience teaching object-oriented programming in Ada. *Symposium on Teaching Object Technology*. Santa Barbara, CA, 1996.
55. M. Ben-Ari. Using inheritance to implement concurrency. *Twenty-Seventh SIGCSE Technical Symposium on Computer Science Education*. Philadelphia, PA, 1996, 180–184.
56. M. Ben-Ari. Distributed algorithms in Java. *Second SIGCSE Conference on Integrating Technology into Computer Science Education*. Uppsala, Sweden, 1997, 62–64.
57. M. Ben-Ari. The software factory. *Psychology of Programming Interest Group Tenth Annual Workshop*. Milton Keynes, UK, 1998, 89–91.
58. M. Ben-Ari. Constructivism in computer science education. *Twenty-Ninth SIGCSE Technical Symposium on Computer Science Education*. Atlanta, GA, 1998, 257–261.
59. M. Ben-Ari. Synchronizing multiple clients and servers. *Ada-Europe International Conference on Reliable Software Technologies*. Uppsala, Sweden, Lecture Notes in Computer Science 1411, 1998, 40–51.
60. Y. Persky, M. Ben-Ari. Re-engineering a concurrency simulator. *Third SIGCSE Conference on Integrating Technology into Computer Science Education*. Dublin, Ireland, 1998, 185–188.
61. Y. Tzuya, M. Ben-Ari. A portable implementation of the distributed systems annex in Java. *SIGADA '98: Ada in Context*. Washington, DC, 1998, 204–211.
62. M. Ben-Ari. Bricolage forever! *Eleventh Annual Workshop of the Psychology of Programming Interest Group*. Leeds, UK, 1999, 53–57.
63. M. Ben-Ari, S. Silverman. DPLab: An environment for distributed programming. *Fourth SIGCSE Conference on Innovation and Technology in Computer Science Education*. Cracow, Poland, 1999, 91–94.
64. M. Ben-Ari, Y. Ben-David Kolikant. Thinking parallel: The process of learning concurrency. *Fourth SIGCSE Conference on Innovation and Technology in Computer Science Education*. Cracow, Poland, 1999, 13–16.
65. M. Ben-Ari. Theory-guided technology in computer science. *Fifth International History, Philosophy and Science Teaching Conference*. Como, Italy, 1999.
66. R. Ben-Bassat Levy, M. Ben-Ari, P.A. Uronen. An extended experiment with Jeliot 2000. *First Program Visualization Workshop*. Porvoo, Finland, 2000, 131–140.

67. Y. Ben-David Kolikant, M. Ben-Ari, S. Pollack. The anthropology of semaphores. *Fifth SIGCSE Conference on Innovation and Technology in Computer Science Education*. Helsinki, Finland, 2000, 21–24.
68. E. Sedletsy, A. Pnueli, M. Ben-Ari. Formal verification of the Ricart-Agrawala algorithm. *Foundations of Software Technology and Theoretical Computer Science* New Delhi, India, Lecture Notes in Computer Science 1974, 2000, 325–335.
69. T. Yeshno, M. Ben-Ari. Salvation for bricoluers. *Thirteenth Annual Workshop of the Psychology of Programming Interest Group*. Bournemout, UK, 2001, 225–235.
70. M. Ben-Ari. The argument for design. *Sixth International History, Philosophy and Science Teaching Conference*. Denver, CO, 2001.
71. M. Ben-Ari, N. Myller, E. Sutinen, J. Tarhio. Perspectives on program animation with Jeliot. *Software Visualization: International Seminar*. Dagstuhl Castle, Germany, Lecture Notes in Computer Science 2269, 2002, 31–45.
72. N. Ragonis, M. Ben-Ari. Teaching constructors: A difficult multiple choice. *Sixth Workshop on Pedagogies and Tools for Learning Object-Oriented Concepts, Sixteenth European Conference on Object-Oriented Programming*. Malaga, Spain, 2002.
73. M. Ben-Ari, N. Ragonis, R. Ben-Bassat Levy. A vision of visualization in teaching object-oriented programming. *Second Program Visualization Workshop*. HornstrupCentret, Denmark, 2002, 83–89.
74. M. Ben-Ari. From theory to experiment to practice in CS education. *Kolin Kolistelut - Koli Calling: Second Annual Finnish/Baltic Sea Conference on Computer Science Education*. Koli, Finland, October, 2002.
75. M. Ben-Ari. Situated learning in this high-technology world. *Seventh International History, Philosophy and Science Teaching Conference*. Winnipeg, Canada, 2003.
76. A. Tikvati, M. Ben-Ari, Y. Ben-David Kolikant. Virtual trees for the Byzantine Generals algorithm. *Thirty-Fifth SIGCSE Technical Symposium on Computer Science Education*, Norfolk, VA, 2004, 392–396.
77. A. Moreno, N. Myller, E. Sutinen, M. Ben-Ari. Visualizing programs with Jeliot 3. *Conference on Advanced Visual Interfaces*, Gallipoli, Italy, 2004, 373–376.
78. M. Ben-Ari, J. Sajaniemi. Roles of variables from the perspective of computer science educators. *Ninth SIGCSE Conference on Innovation and Technology in Computer Science Education*. Leeds, UK, 2004, 52–56.
79. S. Pollack, M. Ben-Ari. Selecting a visualization system. *Third Program Visualization Workshop*. Warwick, UK, 2004, 121–126.
80. C. Yehezkel, M. Ben-Ari, T. Dreyfus. Inside the computer: Visualization and mental models. *Third Program Visualization Workshop*. Warwick, UK, 2004, 77–80.
81. C. Yehezkel, M. Ben-Ari, T. Dreyfus. Computer architecture and mental models. *Thirty-Sixth SIGCSE Technical Symposium on Computer Science Education*, St. Louis, MO, 2005, 101–105.
82. N. Ragonis, M. Ben-Ari. On understanding the statics and dynamics of object-oriented programs. *Thirty-Sixth SIGCSE Technical Symposium on Computer Science Education*, St. Louis, MO, 2005, 226–230.
83. M. Ben-Ari. Whose final hour? The rise of naive egocentric catastrophism. *Eighth International History, Philosophy and Science Teaching Conference*. Leeds, UK, 2005.

84. S. Schwarz, M. Ben-Ari. Why don't they do what we want them to do? *Eighteenth Psychology of Programming Group Workshop*. Brighton, UK, 2006, 266–274.
85. G. Ebel, M. Ben-Ari. Affective effects of program visualization. *Second International Computing Education Research Conference*. Canterbury, UK, 2006, 1–5.
86. R. Ben-Bassat Levy, M. Ben-Ari. We work so hard and they don't use it: Acceptance of software tools by teachers. *Twelfth SIGCSE Conference on Innovation and Technology in Computer Science Education*. Dundee, UK, 2007, 246–250.
87. M. Armoni, N. Lewenstein, M. Ben-Ari. Teaching students to think nondeterministically. *Thirty-Ninth SIGCSE Technical Symposium on Computer Science Education*, Portland, OR, 2008, 4–8.
88. M. Ben-Ari. The effect of the Jeliot animation system on learning elementary programming. *Fourth Greek Conference on the Didactics of Informatics*. Patras, Greece, March, 2008.
89. R. Ben-Bassat Levy, M. Ben-Ari. Perceived behavior control and its influence on the adoption of software tools. *Thirteenth SIGCSE Conference on Innovation and Technology in Computer Science Education*. Madrid, Spain, 2008, 169–173.
90. R. Ben-Bassat Levy, M. Ben-Ari. A survey of research on the Jeliot program animation system. *Fourth Annual Chais Conference on Instructional Technologies Research: Learning in the Technological Era*, 2009.
91. R. Taub, M. Ben-Ari, M. Armoni. The effect of CS Unplugged on middle-school students' views of CS. *Fourteenth Conference on Innovation and Technology in Computer Science Education*, Paris, France, 2009, 99–103.
92. M. Ben-Ari. Teaching concurrency and model checking. *16th International SPIN Workshop on Model Checking of Software*, Grenoble, France, 2009, 6–11.
93. O. Meerbaum-Salant, M. Armoni, M. Ben-Ari. Learning computer science concepts with Scratch. *6th International Computing Education Research Conference*, Århus, Denmark, 2010, 69–76.
94. O. Meerbaum-Salant, M. Armoni, M. Ben-Ari. Habits of programming in Scratch. *Sixteenth Conference on Innovation and Technology in Computer Science Education*, Darmstadt, Germany, 2011, 168–172.
95. J. Lönnberg, M. Ben-Ari, L. Malmi. Java replay for dependence-based debugging. *Workshop on Parallel and Distributed Systems: Testing, Analysis, and Debugging (PADTAD-IX)*, Toronto, ON, 2011, 15–25.
96. J. Lönnberg, M. Ben-Ari, L. Malmi. Visualising concurrent programs with dynamic dependence graphs. *6th IEEE International Workshop on Visualizing Software for Understanding and Analysis*, Williamsburg, VA, 2011, 1–4.
97. J. Lönnberg, L. Malmi, M. Ben-Ari. Evaluating a visualisation of the execution of a concurrent program. *11th Koli Calling: International Conference on Computing Education Research*, Koli, Finland, 2011, 39–48.
98. R. Taub, M. Armoni, M. Ben-Ari. The contribution of computer science to learning computational physics. *6th International Conference on Informatics in Schools: Situation, Evolution and Perspectives*, Oldenburg, Germany, LNCS 7780, pp. 127–137, 2013.
99. M. Ben-Ari. LearnSAT: A SAT solver for education. *16th Int. Conf. on Theory and Applications of Satisfiability Testing*, Helsinki, Finland, 2013, 403–407.

100. R. Taub, M. Armoni, M. Ben-Ari. Abstraction as a bridging concept between computer science and physics. *9th Workshop in Primary and Secondary Computing Education*, Berlin, Germany, 16–19 2014.
101. S. Magnenat, J. Shin, F. Riedo, R. Siegwart, M. Ben-Ari. Teaching a core CS concept through robotics. *Nineteenth Conference on Innovation and Technology in Computer Science Education*, Uppsala, Sweden, 2014, 315–320.
102. S. Magnenat, M. Ben-Ari, S. Klinger, R. W. Sumner. Enhancing robot programming with visual feedback and augmented reality. *Twentieth Conference on Innovation and Technology in Computer Science Education*, Vilnius, Lithuania, 2015, 153–158.
103. R. Ben-Bassat Levy and M. Ben-Ari. Robotics activities—Is the investment worthwhile? *Eight International Conference on Informatics in Schools*, Ljubljana, Slovenia, 2015, 22–31.
104. F. Kaloti-Hallak, M. Armoni, M. Ben-Ari. Students' Attitudes and Motivation During Robotics Activities. *Tenth Workshop in Primary and Secondary Computing Education*, London, 2015.
105. M. Friebroon-Yesharim, M. Ben-Ari. Teaching Robotics Concepts to Elementary School Children. *8th International Conference on Robotics in Education*, Sofia, Bulgaria, 2017, 26–28.
106. R. Ben-Bassat Levy, M. Ben-Ari. The Evaluation of Robotics Activities for Facilitating STEM Learning, *8th International Conference on Robotics in Education*, Sofia, Bulgaria, 2017, 132–137.

Pedagogical Software Tools

Download from GitHub at <https://github.com/motib>, except for the Scratch projects which are at <https://scratch.mit.edu/users/MotiB>. Tools that I developed by myself except for 109.

107. *CP: A Concurrency Simulator*, 1974–80, 1996–2000.
108. *DAJ: Interactive Learning of Distributed Algorithms*, 2001, 2005.
109. (Academic supervision with Erkki Sutinen, University of Joensuu) Niko Myller, Andrés Moreno García, Ronit Ben-Bassat Levy, Roman Bednarik, Noa Ragonis. *Jeliot: A Program Animation System for Java*, 1999–2009.
110. *jBACI: Development Environment and Simulator for Concurrency*, 2000–04.
111. *jSpin: Development Environment for the Spin Model Checker*, 2004–07.
112. *VN: Visualizing Nondeterminism*, 2006.
113. *The Erigone Model Checker*, 2009–11.
114. *Scratch projects: Karel the Robot, Valentino the Robot, Computer Science Unplugged*.
115. *LearnSAT: A DPLL/CDCL/NCP SAT Solver in Prolog*, 2012–17.
116. *Bagrut: Solutions for Israeli Matriculation Examinations*, 2022.
117. *Mosteller's Challenging Problems in Probability*, 2023.
118. *Markov Chain Simulations*, 2023.